Credit Risk
Probability of Default Model

WoE / IV Feature Selection | Logistic Regression | Random Forest
IFRS9 Stage 1 / 2 / 3 | Expected Credit Loss Estimation

0.689	0.378	261,210	$793M	20.1%
AUC-ROC	Gini Coeff.	Loans Scored	Total ECL	Default Rate

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1. Executive Summary

This report presents a complete Probability of Default (PD) credit risk modelling pipeline built on the LendingClub loan dataset (2.2 million records, 1.2 GB). The project develops an end-to-end analytical framework encompassing data preprocessing, industry-standard WoE/IV feature selection, machine learning model development, IFRS9 regulatory staging, and Expected Credit Loss (ECL) estimation.

The primary objective is to demonstrate applied credit risk analytics capabilities aligned with Decision Analytics roles at firms including EXL Service, Deloitte, EY, KPMG, and banking institutions operating under Basel III / IFRS9 frameworks.

0.6889	0.3779	261,210	$793M	20.1%
AUC-ROC (Random Forest)	Gini Coefficient	Loans Scored (Test Set)	Total ECL Provision	Portfolio Default Rate

2. Problem Statement

Credit risk represents the most significant risk category for retail banks. The Basel III accord and IFRS9 standard require institutions to model expected credit losses proactively. The three core components are:

- **Probability of Default (PD)** — Likelihood a borrower fails to meet obligations within a specified time horizon.
- **Loss Given Default (LGD)** — Fraction of the outstanding balance lost upon default (assumed 45% — Basel II standard for unsecured retail loans).
- **Exposure at Default (EAD)** — Outstanding loan balance at time of default.
- **Expected Credit Loss (ECL = PD x LGD x EAD)** — Banks must provision this amount on their balance sheet before losses are realised.

3. Dataset Overview

The LendingClub Loan Dataset (2007-2018) from Kaggle provides real-world retail loan data with sufficient volume and feature richness for a production-grade credit model.

Attribute	Detail
Source	LendingClub via Kaggle (wordsforthewise/lending-club)
Raw size	~2.2 million rows · 150+ columns · 1.2 GB CSV
Target variable	loan_status → binary: Charged Off/Default = 1, Fully Paid = 0
Class balance	~80% non-default / 20% default — addressed with SMOTE
Columns loaded	16 selected via usecols — reduces RAM usage ~75%
Final test set	261,210 loans after cleaning and filtering

Key Predictive Features

Feature	Type	IV Score	Business Interpretation

sub_grade	Categorical	0.479	Strongest predictor — fine-grained risk grade A1 to G5
int_rate	Numeric	0.446	High-risk borrowers charged higher rates
grade	Categorical	0.398	LendingClub risk grade A (safest) to G (riskiest)
loan_to_income *	Engineered	0.119	Debt burden relative to income
installment_burden *	Engineered	0.096	Monthly instalment as % of monthly income
dti	Numeric	0.074	Debt-to-income ratio — measures financial stress
loan_amnt	Numeric	0.034	Total loan amount — determines EAD
revol_util	Numeric	0.026	Revolving credit utilisation — financial stress proxy

* Engineered features created during feature engineering step.

4. Methodology

4.1 Data Preprocessing

Step	Action Taken
loan_status filter	Retain only completed outcomes. Drop Current loans — outcome unknown.
Type correction	int_rate stored as string with % suffix — strip and cast to float64.
emp_length missings	Fill with 'Unknown' category. Missingness itself may signal unemployment.
revol_util missings	Impute with column median — robust to outliers.
annual_inc missings	Impute with median grouped by grade — more accurate than global median.
Columns > 40% missing	Drop entirely — insufficient observations to impute reliably.
Class imbalance (SMOTE)	Applied to training set only after split — prevents data leakage.
Feature scaling	StandardScaler applied before Logistic Regression only. Trees are scale-invariant.

4.2 Feature Selection — Weight of Evidence (WoE) and Information Value (IV)

WoE and IV are the industry-standard credit risk feature selection techniques. Unlike tree-based importance scores, WoE maps every feature bin to a log-odds ratio that logistic regression handles natively, while IV provides a single predictive power score per feature that directly drives model input selection.

Concept	Formula / Definition
WoE	ln (% Non-defaults in bin / % Defaults in bin)
IV	SUM of (% Non-defaults - % Defaults) x WoE across all bins
IV < 0.02	Useless — excluded from model
IV 0.02-0.10	Weak predictor — included with caution
IV 0.10-0.30	Medium predictor — include
IV > 0.30	Strong predictor — high priority (sub_grade, int_rate, grade)
Selected	11 features with IV > 0.02 retained for final model

4.3 Model Development

Model 1 — Logistic Regression (Champion / Scorecard Model): Chosen for regulatory interpretability. Coefficients map directly to scorecard points. Under IFRS9, auditors require attribution of each PD estimate to specific feature contributions — logistic regression satisfies this natively.

Model 2 — Random Forest (Challenger Model): 200 decision trees, max_depth=8, used as performance benchmark. Gini improvement of 4.2 percentage points over LR observed — meaningful but insufficient to justify interpretability trade-off in a regulatory context.

5. Model Evaluation Results

Metric	Logistic Regression	Random Forest	Interpretation
AUC-ROC	0.6679	0.6889	Higher is better. 0.5=random, 1.0=perfect.
Gini Coefficient	0.3358	0.3779	2xAUC-1. Banking standard. >0.30 is acceptable.
KS Statistic	~0.29	~0.32	Max separation between default/non-default curves.
Interpretability	High — scorecard	Low — black box	Regulatory preference: Logistic Regression.
IFRS9 suitability	Fully auditable	SHAP required	LR preferred for provisioning calculations.
Training speed	Fast	Moderate	LR trains in seconds on full dataset.

Note on AUC: A score of 0.67-0.69 is realistic and correct for consumer credit on LendingClub data without feature leakage. Self-reported income and unverified application data introduce noise. Inflated AUC scores (>0.85) on this dataset almost always indicate data leakage from the target variable.

6. IFRS9 Staging Framework

IFRS9 (effective January 2018) requires banks to classify every financial instrument into one of three stages based on change in credit risk since origination. PD scores from the Logistic Regression are mapped to stages using the thresholds below. Provision increases significantly at each stage.

STAGE 1	STAGE 2	STAGE 3
PD < 5%	PD 5–20%	PD > 20%
12-month ECL	Lifetime ECL	Full impairment
55 loans	19,444 loans	241,711 loans
ECL: $14,502	ECL: $18.3M	ECL: $775M

ECL Calculation

Component	Detail
PD	Model output — predicted probability of default per loan
LGD	0.45 (45%) — Basel II standard for unsecured retail loans
EAD	loan_amnt — outstanding balance at time of scoring
ECL formula	ECL = PD x LGD x EAD
Stage 1 basis	12-month ECL — low risk, no significant credit deterioration
Stage 2 basis	Lifetime ECL — significant increase in credit risk
Stage 3 basis	Lifetime ECL on impaired basis — borrower in or near default
Total ECL	$793,324,346 across 261,210 scored test loans

7. Power BI Dashboard

A 4-page interactive Power BI dashboard (credit_risk_pd.pbix) was built connecting directly to the 4 output CSVs. A custom IFRS9-aligned colour theme applies semantic colours: green for Stage 1, amber for Stage 2, red for Stage 3 — consistent across all visuals.

Page	Visuals & KPIs
1 — Portfolio Overview	KPI cards: 261K loans, 20.1% default rate, $3.76B EAD. PD histogram by stage. ECL bar chart. Model AUC summary.
2 — Risk Segmentation	IV feature importance chart (sub_grade top at 0.479). Default rate by interest rate band (line chart).
3 — Model Performance	AUC and Gini comparison: LR vs Random Forest. PD score distribution. Feature importance.
4 — IFRS9 Staging	Donut chart: loan count by stage. ECL bar chart ($M). Full summary table: avg PD, total EAD, total ECL, ECL% of EAD.

8. Technology Stack

Tool	Purpose	Tool	Purpose
Python 3.9+	Core language	Pandas	Data manipulation
Scikit-learn	ML models	imbalanced-learn	SMOTE balancing
XGBoost	Challenger model	SHAP	Explainability
Matplotlib/Seaborn	EDA visualisations	Power BI	Dashboard (4 pages)
Tableau Public	Alt. dashboard	Chart.js	Web prediction demo
ReportLab	This report	Git/GitHub	Version control

9. Business Impact

- **Credit decisioning:** PD scores used at origination to approve loans and set risk-based interest rates. Higher PD borrowers charged higher rates to compensate for additional risk.
- **IFRS9 provisioning:** ECL outputs feed directly into balance sheet provisioning. A $793M ECL on a $3.76B portfolio = 21.1% provision ratio, which must be disclosed in audited financial statements.
- **Portfolio monitoring:** Running the model monthly flags loans migrating from Stage 1 to Stage 2, enabling early restructuring before formal default occurs — reducing actual credit losses.
- **Regulatory compliance:** The Logistic Regression scorecard is fully auditable. Each feature's contribution to a PD score can be attributed explicitly — satisfying IFRS9 and Basel III audit requirements.

10. Future Improvements

Enhancement	Description
Survival analysis	Model when default occurs, not just whether (Kaplan-Meier, Cox hazard).
CCAR stress testing	Adverse scenario: unemployment +3%, GDP -2% — measure portfolio ECL change.
XGBoost + SHAP	Challenger model with full individual-level SHAP explainability.
Vintage analysis	Track cohorts by origination quarter to detect macroeconomic patterns.
Flask REST API	Wrap model in /predict endpoint. Web app frontend already structured.
Model monitoring	Population Stability Index (PSI) to detect feature distribution drift.
MLflow registry	Experiment tracking and model versioning for production deployment.

11. Project File Structure

File / Folder	Description
data/raw/loan.csv	Original LendingClub CSV — never modified
data/processed/scored_loans.csv	261,210 test loans with PD scores and IFRS9 stage
data/processed/ifrs9_stage_summary.csv	3-row IFRS9 summary: loan count, avg PD, ECL per stage
data/processed/iv_scores.csv	IV scores for all 18 candidate features
notebooks/Credit_RIsk_PD.ipynb	Full pipeline: EDA, features, models, IFRS9, exports
dashboard/credit_risk_pd.pbix	Power BI Desktop workbook — 4 pages
website/index.html	Live web demo — PD prediction form with Chart.js visuals
report/Credit_Risk_PD_Report.pdf	This document

12. References

- LendingClub Loan Data (2007-2018). Kaggle. kaggle.com/datasets/wordsforthewise/lending-club

GitHub: github.com/ManishAttry/Credit_Risk_PD | **Live Demo:** manishattry.github.io/credit_risk.html | **LinkedIn:** linkedin.com/in/manish-kumar-mba